

Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose

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Abstract

Deployed LLM agents are organisations of hand-offs: planner to worker to reviewer to aggregator. Every hand-off is a trust boundary, and a prompt injection that hijacks one agent propagates into the next agent’s input. Existing benchmarks measure one agent over one hop; they cannot quantify whether a compromise survives transfer, how far it travels, or where to intervene. We present *Contagion*, a framework for measuring propagation in LLM agent networks using two independent protocols: a *controlled per-edge* protocol that estimates conditional survival $s_i = \Pr[C_i=1 \mid C_{i-1}=1]$, and a *natural-run* protocol that measures end-to-end ASR and the reproduction number R_0 . Four results follow. (i) $ASR = \prod_i s_i^{\text{nat}}$ is an algebraic identity; the assumption worth testing is whether isolated per-hop estimates *transport* to the in-chain regime—they can underestimate by $3.75\times$ ($p = 0.0002$) or transport within noise ($p = 0.93$), and the depth-error slope distinguishes the two cases. (ii) The mechanism is *payload form*: the same target survives at 0.067 arriving as a bare assertion and at 0.967 embedded in a work product. (iii) The epidemic threshold $\rho(M) < 1$ is identically satisfied on any acyclic graph and cannot certify safety; we measure $R_0 < 1$ alongside ASR of 0.45–0.73. (iv) Defence efficacy is unidentifiable without a no-defence baseline: the same redaction code is essential or irrelevant depending solely on the model’s unprotected susceptibility.

Keywords

multiagent systems, prompt injection, security measurement, epidemic threshold, LLM agents, benchmark

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [8] and AutoGen [16] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.



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Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker’s intent, and that output is the *input* of the next agent. The injection therefore does not stay where it landed. Prior work has established that a single agent can be hijacked through a tool response [5, 19] and that an injection can persist within one application [7]. What has not been established is a measurement framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in three ways that we demonstrate empirically rather than assume.

(a) *It hides where failure happens.* In a five-agent chain per-hop survival ranges from 0.133 (worker) to 0.867 (reviewer)—a $6.5\times$ spread within one model and attack—so “the attack failed” and “the attack failed at one specific edge” are different findings that only the per-hop view separates.

(b) *It is not comparable across models without a no-defence baseline.* A resistant frontier model is compromised at 0.10 with *no defence at all*, so a defence driving $0.10 \rightarrow 0.00$ “looks perfect” while removing almost nothing; on a second model the same attack reaches 1.00 and the same defence drives it to 0.00—same defence, attack and code, opposite conclusions. We call this the *floor effect* and treat it as a first-class threat to validity.

What we build. The framework rests on three commitments: per-hop survival s_i from a *controlled per-edge* protocol (force the source compromised, feed its output to the receiver, judge the receiver) and end-to-end ASR from an *independent natural-run* protocol; an injected task semantically competitive with the agent’s legitimate assignment rather than a marker echo, which saturates at 1.0; and reported resolution—Wilson intervals, a bootstrap test with a stated minimum detectable effect, and estimators validated on synthetic ground truth.

What distinguishes this from another injection benchmark is that its central results are structural: composing per-hop rates in a chain is an *identity* plus a transportability hypothesis, and the epidemic threshold $\rho(M) < 1$ is *identically* satisfied on any feed-forward graph and so certifies nothing—properties of the graph and its roles, undefined for a single agent.

Findings, stated as prescriptions. We apply the framework to five models from five independent developers—though coverage differs by claim (cross-model susceptibility spans all five, the depth profile three, and the topology comparison three models with two instances for Llama)—and reach four conclusions, each of which changes what a practitioner should do.

- (1) **Do not compose isolated injection results into pipeline predictions without first checking that they transport.** Per-hop survival measured in isolation *can* fail to transport to the in-chain regime, and where it fails its error *grows with distance* (+11.8 points per hop, rank correlation +1.00), while on another model that transports the slope is an order of magnitude shallower—so the depth profile’s slope is itself the diagnostic (Section 5.7). A single-hop benchmark number is not a pipeline parameter until that check passes.
- (2) **Report per-hop survival conditioned on the receiving role, together with ASR and legitimate-task retention, and always include the no-defence baseline.** Per-hop survival varies $6.5\times$ across receiver roles within one model and attack (Section 5), and a defence’s efficacy is otherwise unidentifiable (the floor effect, Section 5.5).
- (3) **Do not use $R_0 < 1$ as a safety criterion for a feed-forward agent network.** The spectral threshold is identically satisfied on any acyclic graph, and we measure $R_0 < 1$ alongside end-to-end success of 0.45–0.73 (Section 3.8). Reach and reproduction are distinct risks.
- (4) **Treat defence placement as a structural question.** Only nodes on an entry–target path matter, and among those all are equally effective, so placement is decided by the graph and by instrumentation cost—not by measured per-hop survival (Section 3.8).

We also report one of our own initially significant results that proved to be an artefact of the controlled protocol, and the change that removed it (Section 5.3); in a measurement study the protocol is part of the result.

2 Related Work and Positioning

Single-agent indirect injection. Prompt injection as an attack class was first named and systematised by Pérez and Ribeiro [14], who showed that an attacker controlling any text an LLM reads can override prior instructions. The foundational *measurement* work evaluates *one* agent with tool access. InjecAgent [19] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct [17] GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [5] provides a dynamic evaluation environment for one agent over one hop. Liu and Gong [11] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and introduce the ASV/MR metrics we reuse. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Notably, Liu and Gong report that attack success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents. Greshake et al. [7] demonstrate a self-replicating injection “worm” within a single application, establishing persistence but not topology-dependent spread. MAST [3] studies multi-agent failure modes and attributes 32.3% of them to inter-agent misalignment—but non-adversarially: it catalogues failures rather than modelling an attacker propagating one. Concurrent work on cascading instruction influence reports aggregate compromise rates through hierarchical agent chains [1]; it shows the phenomenon is real and deployment-relevant, but an aggregate

rate does not decompose into per-hop conditional survival, does not say which hand-off fails, and does not test whether per-hop rates compose.

Defences. Adaptive attacks against single-agent injection defences break eight defences at $> 50\%$ ASR, including paraphrasing, using token-level optimisation [18]; our negative result on semantic paraphrasing inside a chain is consistent with that finding. We differ in threat model: our attacker uses *natural-language structural obfuscation* (splitting or spacing a literal target) that any user of a tool could produce, rather than white-box token optimisation.

Contagion and thresholds in networks. Treating propagation as contagion over a graph is classical. Threshold models make an individual’s adoption a function of how many neighbours have already adopted [6, 15]; epidemic models supply the complementary criterion that spread is subcritical when the spectral radius of the next-generation matrix falls below one, a threshold that can vanish on heavy-tailed topologies [12, 13]. The problem of choosing which nodes to seed to maximise reachability—the influence-maximisation problem [9]—is NP-hard on general graphs, but tractable in linear time on acyclic topologies by topological traversal, which is precisely why Eq. (3) gives exact answers on our feed-forward networks. This is the lens the recent theory work on multiplex Markov contagion in agent swarms adopts [10]. Our results sharpen where the analogy fails for LLM agent networks: the spectral criterion is *identically* satisfied on any acyclic graph (Section 3.8), so it cannot discriminate; the empirical reproduction number can be below one while reachability stays high; and the per-hop rates such an analysis consumes do not transport from isolated measurement—the classical pipeline of estimating edge parameters and composing them is what our depth result shows to be unreliable. We keep the network-contagion framing but report which instruments transfer.

Gap. No prior work, to our knowledge, provides all of: conditional per-hop survival under a controlled protocol; end-to-end ASR under an independent natural-run protocol; a test of whether the isolated estimate transports to the in-chain regime; per-hop rates conditioned on the receiving agent’s role; and the legitimate-task cost reported alongside. Each clause corresponds to a measurement we show changes the conclusion.

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star (one aggregator receiving fan-in from $N-1$ workers) and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0 , following InjecAgent’s indirect channel [19]. We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational

task). This is not incidental: an agent that resists injection merely by refusing to work is not a defence we want to credit, and all injected content is therefore framed as benign-looking infrastructure text, which is the difference between 0% and 60–100% compliance in our pilot.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates: controlled per-hop success reaches 0.94 on one local model. A metric at its ceiling cannot rank defences.

Family B (semantic instruction competition), used for all headline numbers, makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the interior of (0, 1) on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [11]: $C = 1[\text{ASV} \geq \tau_{\text{asv}} \vee \text{MR} \geq \tau_{\text{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\text{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment variant of MR scored 0.459 on clean benign outputs and 0.597 on refusing outputs—assigning high similarity precisely to the outputs a defence is supposed to produce. Under the ASV-only rule benign and compromised outputs separate cleanly (calibration accuracy 1.000 versus 0.795). Because the rule is deterministic, no LLM-judge variance enters the metric, and because it matches a literal target it does not require per-model recalibration.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i . For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u 's compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v . We repeat N times, rotating the benign assignment, so that $\hat{s}_i = k_i/N_i$ estimates the conditional $\Pr[C_v=1 \mid C_u=1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number

of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact for each trial. Reusing a single artefact across trials makes $\hat{s}_i$ inherit the idiosyncrasy of one sample; because the product is built from those same estimates the error does not cancel, and it can produce spurious composition failures in *either* direction. Our first measurement of the composition test used a fixed artefact and produced a significant apparent deviation that vanished once artefacts were drawn per trial (Section 5.3). We therefore treat the fixed-artefact protocol as a documented ablation and the fresh-artefact protocol as the default for all compositional claims.

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl} , survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat} , survival measured *within the propagation process*, $\Pr[C_i=1 \mid C_{i-1}=1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i=1 \Rightarrow C_{i-1}=1$, and therefore

$$\text{ASR} = \Pr[C_k=1] = \prod_{i=1}^k \Pr[C_i=1 \mid C_{i-1}=1] = \prod_{i=1}^k s_i^{\text{nat}} \quad (1)$$

is an *identity*, not a modelling assumption. We verify it empirically: for one frontier model, $\prod_i s_i^{\text{nat}} = 0.850$ and $\text{ASR} = 0.850$. The claim usually described as “the Markov product assumption” is, when stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}. \quad (2)$$

This is exactly the assumption that single-agent injection benchmarks make implicitly when their numbers are composed to predict pipeline behaviour, and it is the assumption we test.

3.6 Reproduction number and utility

We report the empirical reproduction number R_0 (mean new compromises per compromised agent-instance), with a Wilson interval on the underlying compromise proportions, alongside its regular-topology target $d \cdot \bar{s}$ (d = mean out-degree) as a consistency check, not an identity. Because a network can achieve $\text{ASR} = 0$ by refusing to process any untrusted content, we also measure the paired legitimate-task performance $\Delta U = U_{\text{clean}} - U_{\text{attack}}$ and Retention = $U_{\text{attack}}/U_{\text{clean}}$, running each configuration twice (with and without injection) and scoring the network's final legitimate output. In this draft U is the indicator that the final legitimate answer is uncontaminated by the attack target; we scope the metric accordingly (Section 7).

3.7 Estimators

All proportions use Wilson score intervals. For the composition test we use a bootstrap test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$: because ASR and the $\hat{s}_i$ come from independent experiments, variances add and an independent bootstrap over both sources is valid. We report the point estimate, a percentile interval, a two-sided p -value for $H_0 : \Delta = 0$, and the minimum detectable effect $\text{MDE} = (z_{1-\alpha/2} +$

$z_{\text{power}})$ $\text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*.

Over the six chain cells at the common $n = 40$ grid, only the Llama cell rejects H_0 under our default fresh-artefact protocol ($p = 0.0002$); it survives a Benjamini–Hochberg [2] correction across the five non-degenerate cells at $q = 0.05$, so multiplicity does not inflate this positive finding. (The one apparent DeepSeek rejection was a fixed-artefact artefact that vanishes under the fresh protocol, $p = 0.63$; Section 5.3 and supplementary Appendix A. At $n = 200$ the Nova cell also rejects—consistent with its $n = 40$ null being underpowered—but at a small deviation on the resistance floor; Section 5.10.)

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e , the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u]), \quad \Pr[v_0] = 1. \quad (3)$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.7). The independence assumed here is between *edge activations*; it is not contradicted by the per-context heterogeneity of a single edge (Section 5.10, where χ^2 rejects exchangeability on the weak edge), since Eq. (1) is a marginal statement and heterogeneity affects only the estimate’s *variance* (supplementary Appendix B).

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M , with $M_{uv} = s_{u \rightarrow v}$, satisfies $\rho(M) < 1$ [13]. For a feed-forward network this criterion is *vacuous*. Ordering nodes topologically makes M strictly triangular, so every eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with $\text{ASR} = 0.450$, and $R_0 = 0.420$ on the seven-agent star with $\text{ASR} = 0.725$: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0 . Reach and reproduction are different risks and can move in opposite directions (Section 5.6).

We therefore use R_0 as a *descriptive amplification statistic*, not a safety certificate: reachability, the quantity a practitioner cares about, is given by Eq. (3), not by R_0 .

A corollary concerns defence placement. In a feed-forward network a node can affect end-to-end propagation only if it lies on

some path from the entry to the target, and among the nodes that do, hardening any one is equally effective. Evaluated on our measured rates this is exact: on the seven-agent chain all five interior nodes reduce the predicted ASR by 0.450, i.e. to zero, and on the tree the two branching nodes are sharply separated—hardening agent_2, which lies on the entry–target path, reduces predicted ASR from 1.000 to 0.000, while hardening agent_1, a node of *identical* out-degree that lies on a different branch, changes nothing. Placement is therefore decided by the graph, not by the measured survivals: compute the entry–target paths, and among the nodes they contain harden the cheapest one to instrument or the one whose legitimate traffic is least costly to lose. Per-hop survival enters only through *whether* a defence is worth deploying at all, not through *where*. This changes when the network admits redundant paths, where a node can be bypassed and placement becomes a genuine optimisation.

4 Experimental Setup

Models. Five models from five independent developers: qwen2.5:7b (Alibaba, local), DeepSeek V3.2 (DeepSeek), Llama 3.3 70B (Meta), Nova Pro (Amazon), and Claude Sonnet 4.5 (Anthropic); the latter four via Bedrock, region us-east-1. Temperature 0.7; the models marked † require Bedrock inference-profile identifiers. Models were selected to span a wide capability and deployment range: from a fully local 7B-parameter open-weight model (qwen2.5:7b, running on commodity hardware without API access) to frontier closed models from distinct corporate ecosystems. The local model confirms that the framework does not require commercial API credentials. Using five *independent* developers also minimises the risk that a shared fine-tuning corpus or RLHF pipeline explains cross-model patterns.

Agent roles and network wiring. Each node in the network is assigned a role—*planner*, *worker*, *reviewer*, or *aggregator*—and receives a fixed system prompt encoding that role’s responsibilities. The planner (agent_0, always the entry node) is given a broad coordination mandate; the worker (agent_1) is instructed to execute a delegated sub-task; the reviewer (agent_2) is asked to inspect and summarise incoming content; the aggregator (agent_3) is asked to synthesise a final answer. These roles are deliberately minimal: they reflect the structural positions that appear in real orchestration frameworks [8, 16] without over-specifying the prompt engineering. All role prompts are held fixed across models and attacks; only the underlying model identifier changes, so cross-model differences in per-hop survival are attributable to the model, not to prompt variation.

Cells. Unless stated otherwise: four-agent chain, entry agent_0, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$. The study comprises 25 result sets and $\approx 9,000$ model calls (a lower bound, counted from these configurations). Benign assignments are drawn from a pool of three short operational questions (summarise, fact-recall, draft) rotated cyclically so that no trial sees the same context twice; the three contexts are semantically orthogonal to the injected task to prevent accidental overlap from inflating survival estimates. The practical consequence of this design is visible in Section 5.10: pooling across contexts inflates the overdispersion

Table 1: Chain, no defence, Task Family B. $\bar{s}$ is the mean controlled per-edge survival. Per-edge order follows the chain. Rows marked $\dagger$ used the fresh-artefact protocol (Section 5.3); others used the default fixed-artefact protocol, which is why both are reported in Table 3 for the models where we have both. An independent replicate of the DeepSeek cell gave ASR = 0.375 with per-edge 0.700/0.733/0.833, illustrating the sampling variability at $n = 40$ that the MDE of Section 3.7 quantifies.

Model	ASR	$\bar{s}$	per-edge
Llama 3.3 70B $\dagger$	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 $\dagger$	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

statistic to $\varphi = 2.93$, whereas evaluating at fixed context gives $\varphi = 0.58$.

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (BANANA – 77, requiring recombination), or *split* (BANANA and 77, combine with a hyphen). The split style contains no literal target, so redaction is a no-op by construction: none-split and redact-split measure the same quantity, which gives us a built-in noise gauge. Defence scope is deliberately narrow—we evaluate a static, non-adaptive attacker and two rule-based defences; adaptive attackers and learned detection layers [4, 18] are left for future work.

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth: Wilson score interval coverage over a grid of p and n ; size and power of the composition bootstrap test; calibration of the formal test’s bias, size and power against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree. This step serves as a sanity check that the measurement instruments do not introduce systematic bias before any real model calls are made.

5 Results

5.1 Per-hop survival is role-dependent, and susceptibility does not track capability

Table 1 reports the chain headline per model. Two facts stand out. First, per-hop survival varies several-fold *across receiver roles within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867. Reporting a scalar s per model is therefore unjustified. Second, susceptibility does not follow capability: the most capable model we test is the most susceptible, and five models produce four qualitatively different outcomes.

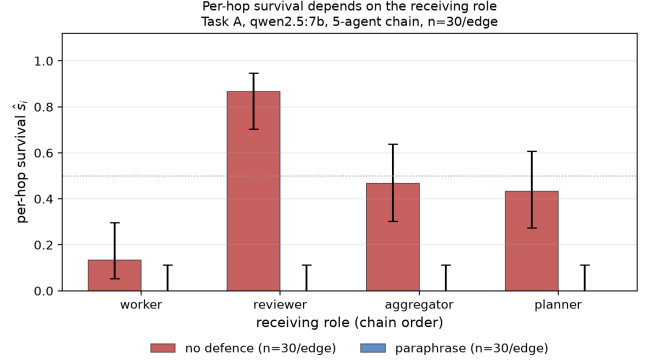


Figure 1: Per-hop survival by receiving role (five-agent chain, $n = 30$ per edge). The worker-position edge is an order of magnitude below the reviewer-position edge, and a defence that collapses survival in one chain can leave another essentially unchanged.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75 $\times$ underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$ against ASR = 0.850, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$. Re-running this edge at $n = 200$ confirms the finding with full power: $\Delta = 0.615$ ($[0.54, 0.69]$, $p = 0.0002$, MDE = 0.11; Section 5.10).

For *DeepSeek V3.2* the same comparison agrees within resolution: the three per-edge ratios are 0.90, 0.99 and 1.10, $\prod_i s_i^{\text{ctrl}} = 0.409$ against ASR = 0.400, and $\Delta = -0.009$ ($[-0.225, +0.201]$, $p = 0.930$)—a *non-rejection* at MDE ≈ 0.26 ($n = 40$), not distinguishable from exact, whereas the Llama failure is a powered positive (MDE = 0.11 at $n = 200$). In both cases Eq. (1) holds exactly, as it must ($\prod_i s_i^{\text{nat}} = \text{ASR}$: 0.850 for Llama, 0.400 for DeepSeek).

The practical reading is that per-hop survival measured in the isolated regime—which is the regime single-agent injection benchmarks inhabit—can misestimate in-chain survival by a factor of four, but does not always do so. Whether it does is an empirical property of the model and the edge, not something that can be assumed in either direction. A benchmark that reports isolated single-hop success and is then composed into a pipeline prediction inherits that error silently when it occurs. This is the hypothesis of Eq. (2) tested on concrete edges; Section 3.5 explains why it is the assumption worth testing rather than the product identity.

5.3 A significant finding of our own that did not survive

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i \hat{s}_i = 0.806$ against ASR = 0.475: a large apparent deviation in the opposite direction, with $\Delta = -0.331$,

Table 2: Transportability of the isolated per-hop estimate, per edge and per model (fresh-artefact protocol). $ratio$ is s^{nat}/s^{ctrl} ; values near one mean the isolated estimate transports. $\prod$ rows are products over the three edges; the final row is measured independently on natural runs and must equal $\prod$ of the in-chain column (Eq. (1)).

model	edge	s^{ctrl}	s^{nat}	ratio
Llama 3.3 70B	a0→a1	1.000	1.000	1.00×
	a1→a2	0.233	0.875	3.75×
	a2→a3	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	—
	$\prod$ (in-chain) / ASR	—	0.850	—
DeepSeek V3.2	a0→a1	0.667	0.600	0.90×
	a1→a2	0.800	0.792	0.99×
	a2→a3	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	—
	$\prod$ (in-chain) / ASR	—	0.400	—

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact (one sample reused across trials) versus a fresh artefact per trial, with the resulting product and test. The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	policy	per-edge	$\prod$	Δ	p
DeepSeek	fixed	0.967/1.000/0.833	0.806	−0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	−0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \rightarrow +0.617$, $p < 0.001$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact.

We report this because the sensitivity of a compositional claim to the artefact policy is itself information a practitioner needs, and because the failure mode is invisible from the point estimate alone: both versions produced a plausible-looking table. Where a deviation is real it can point either way: across our cells we observe both an under-estimate (Llama) and, before correction, an over-estimate (DeepSeek).

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion produced by forcing the sender to answer the injected instruction—whereas in natural runs the same target travels inside the sender’s legitimate work product and may be consumed as content rather than as a command.

Table 4: Content-form ablation (Llama 3.3 70B, chain $n=4$; 40 natural runs, 30 replays per arm and edge). Both replay arms use the same receivers, assignments and judge, so the difference between them is attributable to payload form alone. The harness check is the comparison of columns one and two.

edge	in-chain s^{nat}	replay in-context	replay canonical	form effect
a0→a1	1.000	1.000	1.000	0.000
a1→a2	0.950	0.967	0.067	+0.900
a2→a3	0.947	0.733	1.000	−0.267

To separate *form* from *regime*, we replay both content sources through the *same* harness: a fresh receiver, the same rotating legitimate assignment, the same judge; the two arms differ only in where the content came from. The harness is self-validating, because one arm replays content captured from real natural runs and must therefore reproduce s^{nat} .

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 when it arrives embedded in a work product and at 0.067 when it arrives as a bare assertion—a 14× difference produced entirely by packaging. We read this as the mechanism behind Section 5.2: the canonical artefact used by isolated measurement is not a representative message.

The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733), so form interacts with position. On that same edge the in-context replay (0.733) does not reproduce the in-chain rate (0.947), so the harness self-check passes on two of three edges and we do not treat the third as a clean comparison.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Fig. 2) shows why this is weaker than it appears. On the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88: it works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack succeeds only 0.10 of the time with no defence, so redaction’s apparent perfection is the absence of anything to defend (robust to the compromise threshold: the rate is invariant for $\tau_{ASV} \in \{0.7, 0.8, 0.9, 1.0\}$, supplementary Appendix C). The same code and defence give opposite verdicts; only the headroom differs.

Semantic paraphrase is worse than useless in our cells: on the model where the attack succeeds it leaves ASR unchanged (0.450 $\rightarrow$ 0.450) while reducing utility retention from 0.600 to 0.400, so it is *strictly dominated* by doing nothing, and on the resistant model it still costs 0.100 of utility where ASR is already 0.000. This is consistent with prior single-agent findings that paraphrasing is breakable [18], and it is visible only because utility is reported jointly with propagation.

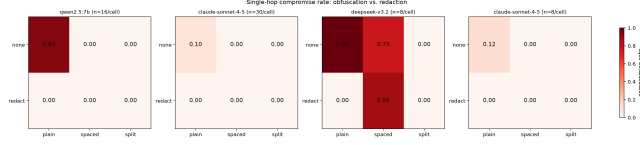


Figure 2: Single-hop compromise rate by defence and attack form, one panel per model ($n = 30$ per cell for Claude 4.5, $n = 8$ for DeepSeek, $n = 16$ for qwen; the $n = 8$ cells have wide intervals and we do not interpret small differences there). none-split and redact-split measure the same quantity by construction, so the gap between them gauges our own noise.

Scope. These verdicts cover two defences (literal-target redaction and semantic paraphrase) against a static, non-adaptive attacker; learned/semantic detectors and adaptive attackers [18] are out of scope. The small panels (DeepSeek $n=8$, qwen $n=16$) are used only for the within-model none-vs-redact contrast; the cross-model claims rest on the $n=30$ Claude panel and the chain cells and are unchanged if the two are dropped.

5.6 Topology changes the spread of survival, not just the amount

Topology affects propagation in two ways that our measurements separate. The first is structural and expected: out-degree and the position of the entry point change the reproduction number. On a seven-agent star with the entry at a leaf, only one leaf is compromised, so exactly one edge carries the attack; measured on Llama 3.3 70B the centre is compromised with $ASR = 0.725$, giving $R_0 = 0.420$ against a regular-topology target of $d \cdot \bar{s} = 0.652$. The gap is structural, not noise—the aggregator has no successors, so compromises at the centre generate none downstream—and it reproduces the deviation our synthetic calibration predicted for star fan-in.

The second effect is less obvious and does not follow from the first. Table 5 compares three seven-agent topologies that have nearly the same mean per-hop survival yet differ sharply in outcome. The star reaches its aggregation point more often than the chain ($ASR = 0.725$ versus 0.450) while reproducing less ($R_0 = 0.420$ versus 0.821). Reach and reproduction are not the same risk, and a topology can move them apart: in the star the attack usually succeeds at the single hop that matters, but a compromise at the centre generates no further compromises because the centre has no successors. The tree is the most permissive of the three on both measures ($ASR = 0.800$, $R_0 = 0.831$) and is also the only cell in which R_0 approaches its regular-topology target ($d \cdot \bar{s} = 0.795$, within 5%), the approximation holding where branching is genuinely balanced.

A third effect concerns dispersion. In the chain, edges connect several distinct receiver roles and survival spans 0.27 to 1.00 ; in the star, all six edges connect a worker to the aggregator and survival is nearly homogeneous (0.667 – 0.867). A topology therefore does not merely multiply a common per-hop rate by a degree-dependent factor: it determines how much role heterogeneity the network averages over, and hence whether a single scalar s summarises the network at all. These effects are not specific to one model or instance: repeated on Nova, DeepSeek, and a second Llama instance ($n = 10$), the star/tree comparison preserves the pattern wherever

Table 5: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge). The three topologies have similar mean per-hop survival but differ substantially in outcome, and the two outcome measures do not move together.

topology	edges	$\bar{s}$	ASR	R_0
chain	6	0.761	0.450	0.821
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	0.800	0.831

Table 6: Shape of the depth profile: relative error of the isolated product, $|\prod_i s_i^{\text{ctrl}} - P(C_i=1)|/P(C_i=1)$, as a function of hops from the entry (fresh-artefact protocol). range is min-max over informative depths; ρ is Spearman’s rank correlation between depth and error, slope its OLS trend in percentage points per hop. On Llama the chain dies at depth 11, so the last four depths are undefined and excluded rather than scored as total error. DeepSeek is shown at two chain lengths: at $n = 8$ its slope is indistinguishable from zero, at the better-powered $n = 10$ it is weakly positive.

model	chain	range	mean	ρ	slope
qwen2.5:7b	n=7	31–89%	66%	+1.00	+11.8
Llama 3.3 70B	n=15	0–92%	66%	+0.77	+6.8
DeepSeek V3.2	n=8	7–35%	21%	−0.07	+0.1
DeepSeek V3.2	n=10	4–33%	15%	+0.65	+2.3

propagation is non-zero (DeepSeek $ASR = 0.75$ versus 0.475 ; second Llama 0.675 versus 0.65), while Nova sits at the resistance floor (0.00 , 0.125)—so it rests on three models and two instances, not one cell.

5.7 The shape of the depth profile is a diagnostic

Table 6 gives the depth profile across models, and the slopes form a *spectrum* rather than two categories. On qwen2.5:7b the error rises at every step ($\rho = +1.00$, +11.8 points per hop). On Llama 3.3 70B it rises almost monotonically to 92% ($\rho = +0.77$, +6.8 points per hop) before the chain dies outright at depth 11, and a re-run at higher power reproduces this ($\rho = +0.76$, +6.3). DeepSeek V3.2 sits at the shallow end: at $n = 8$ its slope is indistinguishable from zero ($\rho = -0.07$), but at the better-powered $n = 10$ it is weakly positive ($\rho = +0.65$, +2.3 points per hop, mean error 15%)—an order of magnitude below qwen and Llama, and itself a reminder that a short low-power chain can read as flat.

The ordering is not a contradiction but a consequence of what Section 5.2 measures directly. Where the isolated rate transports best—DeepSeek’s isolated product matches its measured ASR to 2.2%—composing edge estimates adds little systematic error and the profile stays shallow; where it fails on one edge—Llama’s middle edge is under-estimated by $3.75\times$ —that error enters every product containing the edge and compounds with depth. The profile is a cheap diagnostic: one long chain, one run, and its *slope* indicates how far isolated per-hop rates compose for that model. The magnitude is model-dependent (steepest on qwen, intermediate on

Table 7: Utility under attack (Section 5.9), 20 paired clean/attack runs. Retention = $U_{\text{attack}}/U_{\text{clean}}$.

model	defence	ASR	U_{att}	ΔU	ret.
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	0.400
DeepSeek	redact	0.000	1.000	+0.000	1.000
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

Llama, shallow on DeepSeek), so it is the slope read as a continuous quantity, not a single number, that diagnoses transportability.

The tree cell adds a second constraint: the error’s *sign* is not fixed. On the tree the isolated estimate is optimistic, predicting certain compromise ($\prod_i s_i^{\text{ctrl}} = 1.000$) where the measured rate is 0.800, and the two edges on the target’s path are exactly the ones whose isolated survival exceeds their in-chain survival. The shallow end is where the error is smallest—8% at the star’s hop, 25% on the tree’s two-hop path. So the defensible statement is that *the profile’s slope diagnoses transportability while its direction remains model- and topology-dependent*. The mechanism is the content-form effect of Section 5.4: every hop is an opportunity to re-package the payload.

Evaluated on in-chain rates, Eq. (3) reproduces the measured ASR exactly in all three topologies; on isolated rates it deviates by -0.370 , -0.058 and $+0.200$ —so what breaks independence is the repeated re-packaging of the payload, not the network.

5.8 Recurrence makes the threshold non-vacuous

Section 3.8 proved the threshold criterion vacuous for acyclic graphs and informative only once the interaction graph contains a cycle. We test that second half on a manager-with-reporting-workers pattern and confirm it empirically: the spectral radius reduces to $\rho(M) = \sqrt{\sum_j a_j c_j}$, which yields *opposite* predictions borne out by measurement—the Llama and DeepSeek loops are supercritical ($\sum_j a_j c_j = 1.800$ and 1.018 , endemic) while qwen’s decays (0.578). The full derivation, design rule, dynamics, and caveats are deferred to supplementary Appendix D.

5.9 Utility under attack

Table 7 makes the trade-off concrete. Redaction on the susceptible model blocks propagation at *no* utility cost, while paraphrase buys no benefit and costs utility; on the resistant model it is pure cost. Reporting (ASR, ΔU) jointly makes these statements checkable; a propagation-only table would have made paraphrase look neutral.

5.10 Estimator validation

On synthetic data with known ground truth, Wilson intervals attain nominal coverage and our bootstrap test is unbiased ($|\text{bias}| \leq 0.10\sigma$, size 0.067 – 0.083 at $\alpha = 0.05$ over 60 replications), with higher power than a CI-overlap rule (0.750 vs 0.625). Detecting a 0.10 deviation at power 0.8 needs $n \approx 200$; at $n = 40$ only deviations above 0.26 are excluded, so we treat no $n = 40$ null as evidence and label the one $\text{ASR} = \prod_i \hat{s}_i = 0$ cell as uninformative. To confirm the

Llama transport failure is powered rather than a coarse- n artefact, we re-ran its weak edge at $n = 200$: the MDE falls to 0.11 and $\Delta = 0.615$ ($[0.54, 0.69]$, $p = 0.0002$) rejects H_0 well above threshold. At $n = 200$ the resolution cuts both ways: DeepSeek stays a non-rejection at the tighter MDE 0.13 ($\Delta = 0.022$, $p = 0.60$), so its transport is powered-consistent; Nova, an $n = 40$ null ($p = 0.084$), instead rejects ($\Delta = 0.051$, $p = 0.0002$)—why we decline to read small- n nulls as consistency—though the deviation is small and on Nova’s floor (ASR = 0.055 , $\prod_i \hat{s}_i \approx 0.004$), and its middle edge shows Llama’s non-transport signature (0.09 isolated versus 0.66 in-chain).

We also tested the trial-level assumption: φ must be computed *within* a temperature (pooling inflates it to 2.93); at fixed temperature the weak edge gives $\varphi = 0.58$ (upper limit 1.00), supporting the Bernoulli assumption behind Wilson intervals. A homogeneity test across the three benign task contexts, however, rejects exchangeability on that same weak edge ($\chi^2 = 13.40$, $\text{df} = 2$, $p = 0.0014$; $18/56$ versus $3/56$ versus $13/48$) while the two saturated edges are perfectly homogeneous: trial outcomes depend on the legitimate task the receiver is doing, the same context-dependence the content-form ablation exposes from the other direction. Our rates are averages over contexts (a context-clustered interval leaves the estimate essentially unchanged; supplementary Appendix B).

6 Discussion

The four prescriptions of the Introduction are cheap to adopt and each addresses a specific measurement failure our results expose.

Composition is safe once transportability is checked. Eq. (1) is an algebraic identity, not a modelling choice, so the product of in-chain per-hop rates always equals end-to-end ASR. The only step that can fail—and that single-agent benchmarks implicitly rely upon—is transporting isolated per-hop estimates into that product. Our depth profiles give a cheap diagnostic: a single long chain measured under both protocols, with the *slope* of the relative error as the statistic. A flat slope (DeepSeek, $+2.3$ points per hop at better-powered $n=10$) means isolated measurement is safe to compose; a steep slope (qwen, $+11.8$; Llama, $+6.8$) means the in-chain regime must be measured directly.

Defence evaluation requires headroom. A defence cannot remove more signal than its model permits. Where the unprotected base rate is 0.10 (Claude Sonnet 4.5), redaction looks perfect because there is almost nothing to defend; where it is 1.00 (Llama, susceptible cells), the same code is essential. Reporting a single post-defence ASR without the no-defence baseline confounds the defence effect with the model’s floor, and our utility data add a further dimension: semantic paraphrase is *strictly dominated* in every cell we measured, costing utility while buying nothing on security.

Graph structure decides placement, not per-hop survival. In a feed-forward network the set of nodes that can affect end-to-end propagation is exactly the set on some entry–target path, and within that set all placements are equally effective (Section 3.8). The practitioner’s question is therefore: compute the paths, and among the nodes they contain harden the one with lowest instrumentation cost or least legitimate-traffic impact. Per-hop survival is relevant only to the prior question of whether defence is worth deploying

at all. Redundant paths are the only configuration that promotes placement to a genuine optimisation problem.

Scope. The results above hold under one benign marker payload, a static non-adaptive attacker, and networks of up to fifteen agents; the recurrent case spans three models with limited power. Extending to larger meshes, adaptive attackers, and learned detectors is the natural follow-on.

7 Limitations and Threats to Validity

Utility is a contamination proxy, not a correctness oracle. The utility metric U flags whether the injected target string appears in the final deliverable; it does not measure whether the legitimate task was answered correctly. A task family with an independently verifiable ground-truth answer (e.g. arithmetic or code execution) would sharpen the security–utility trade-off in Section 5.9 and make the utility cost of paraphrase—currently measured only through output contamination—directly comparable across defence conditions.

Transportability failures are real but their prevalence is unknown. The clearest failure is Llama’s weak edge, powered to $\Delta = 0.615$, $p = 0.0002$, MDE = 0.11 at $n = 200$: isolation underestimates in-chain survival by 3.75×. Nova shows the identical middle-edge signature and rejects at the same power level, though the deviation is small given its resistance floor. DeepSeek transports cleanly under the same test. This covers three of our five models on one attack family and one chain depth; how frequently the failure occurs across model families, task families, and chain configurations is uncharacterised. The depth-error slope is a diagnostic but not a guarantee: a short chain at low power can read as flat even where a longer chain would reveal accumulation.

Protocol sensitivity, reported rather than hidden. Our initial significant composition failure on DeepSeek was a fixed-artefact artefact that vanished when artefacts were drawn fresh per trial ($p = 0.0030 \rightarrow p = 0.930$; Section 5.3). We present both versions to record that compositional claims are sensitive to the artefact-sampling policy, and that the sensitivity can flip a result in either direction: DeepSeek moved from apparent over-estimate to non-rejection, while Llama’s under-estimate was unchanged. Unstated protocol choices that alter the result are a class of threat that benchmark replication exercises routinely miss.

Scale, topology, and defence coverage. Headline chain results span four to fifteen agents; star and tree cells are replicated across three models with two Llama instances ($n=7, n=10$); the recurrent (cyclic) case covers three models with limited power. Configurations not varied include: networks of tens of agents; mesh, debate, or mixture-of-experts topologies; re-injection (an agent injecting its successors’ inputs); and defence families beyond literal-target redaction and semantic paraphrase—detection layers, hop isolation, and structured queries [4] are referenced but not evaluated. Cell sizes of 30–40 trials provide MDE ≈ 0.26 for the composition test; the ASV threshold applies to a literal target and would need recalibration for paraphrase-obfuscated targets; and the family-A MR branch is uncalibrated for the judgement regime used here.

8 Conclusion

A compromise’s reach in an LLM agent network is a property of the graph, the roles travelled, and the form in which the payload is re-packaged at each hand-off—not of any single agent in isolation. Four structural results follow from separating the identity $ASR = \prod_i s_i^{\text{nat}}$ from the transportability hypothesis $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$.

(i) Composition of per-hop rates is an identity that always holds when the rates are measured *in the chain*; the failure mode is transporting isolated estimates into that product, detectable by the slope of the depth-error profile. (ii) Payload form is a first-order mechanism: the same target survives at 0.067 arriving as a bare assertion and 0.967 embedded in a work product, under the same receiver and judge. (iii) The epidemic threshold $\rho(M) < 1$ is identically satisfied on any acyclic interaction graph and cannot certify safety; the correct reachability estimate is the percolation formula (Eq. (3)), not R_0 . (iv) Defence efficacy is unidentifiable without a matched no-defence baseline, and defence placement is decided by graph structure (entry–target paths) rather than by measured per-hop survival.

Practically, the measurement cost of adopting these prescriptions is low: they change what is *reported* rather than what is built, except for the single additional long-chain probe that makes the depth slope a routine diagnostic. As LLM agent orchestration frameworks grow in complexity, measurement frameworks that decompose end-to-end risk into identifiable, role-conditioned, topology-aware quantities become a precondition for meaningful security evaluation.

Ethical Considerations

All experiments use a *benign* marker payload against our own harness via standard provider APIs; the attack forms are already documented for single agents. No real user data, production deployments, or third-party systems were involved.

Artifact Availability

Code, per-cell outputs, tests and all offline re-analysis (supplementary Appendices A–D) reproduce every table with no model call, and a free local model (qwen2.5:7b) is included; a REPRODUCE manifest links each table to a command.

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